

ETE4351L Data Communications and Networking

Lab 3 Sampling Simulation

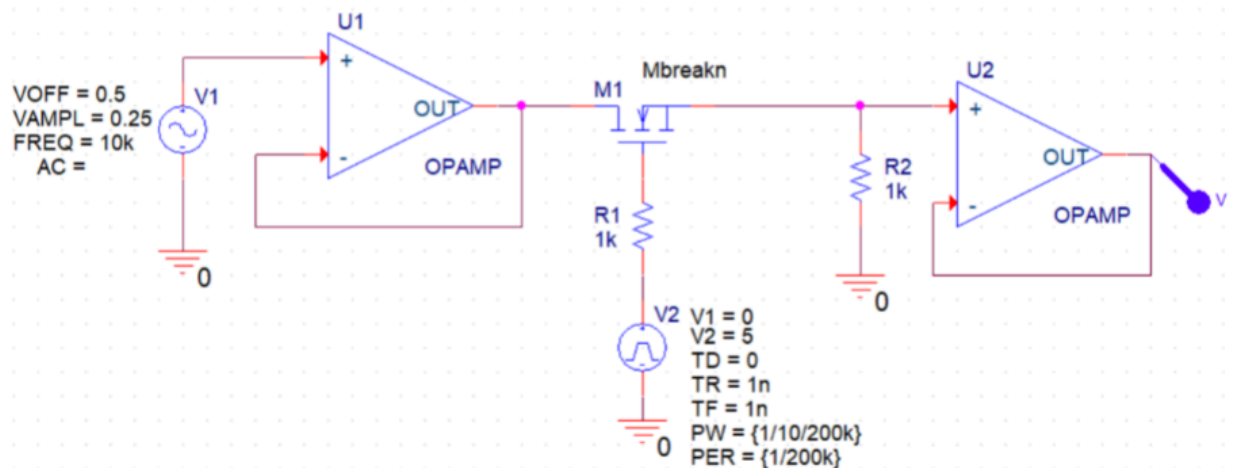
Mariana Rivera, Sarah Boyd, Fatima Saldaña-Gutierrez

February 10, 2026

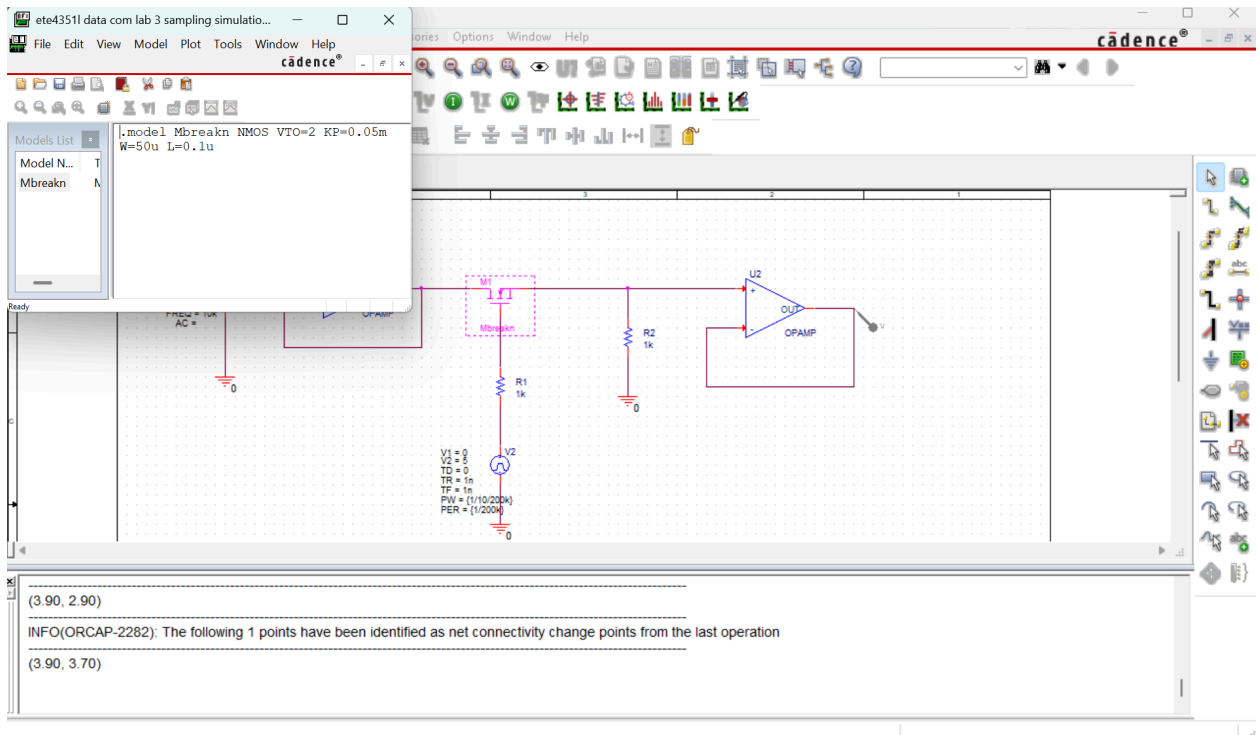
Introduction:

The lab is a simulation on sampling signals at varying frequencies and how each frequency sample compares within the time and frequency domain. The given circuit sets a 0.5V sine wave source with the frequency source at 10kHz, and an offset eliminating negative voltage. The source frequency is placed through a buffer amplifier (to lower impedance and increase current) before entering a NMOS transistor. On the transistor's gate, the sampling frequency is received and will vary throughout this lab. The product between the two signals will pass a pull down resistor and enter the final buffer amplifier. The pulses will outline an envelope in the time domain and the same envelope in the frequency domain. The frequency domain will present the compression and expansion of frequency content. Additional modifications will be made to the base circuit. Switching out the sine wave source with a pulse waveform at 50% duty cycle to observe the modulation between two pulses. Reverting back to the original circuit, a 15nF capacitor is added and the pull down resistor increased from 1kohm to 100kohm. This is intended to create the “holding” effect where the voltage is stable in the sample reading.

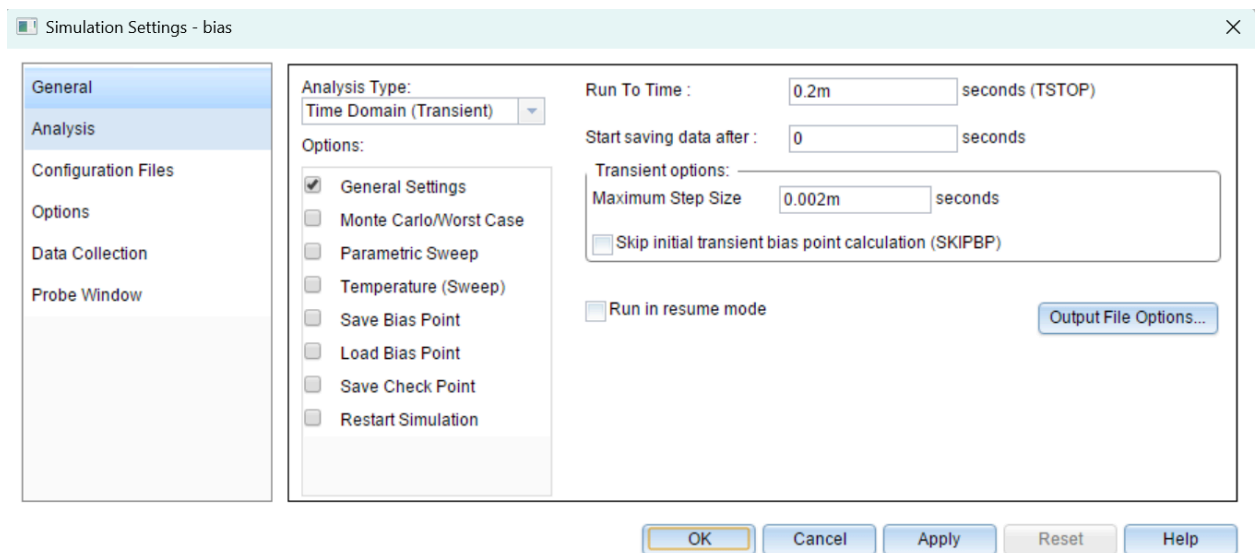
1. Building the circuit



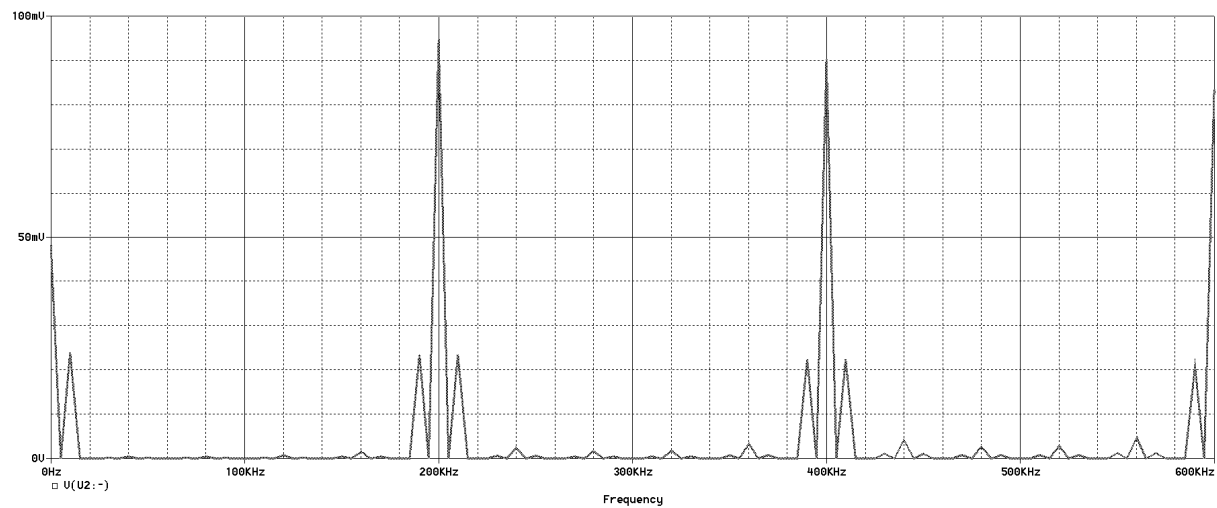
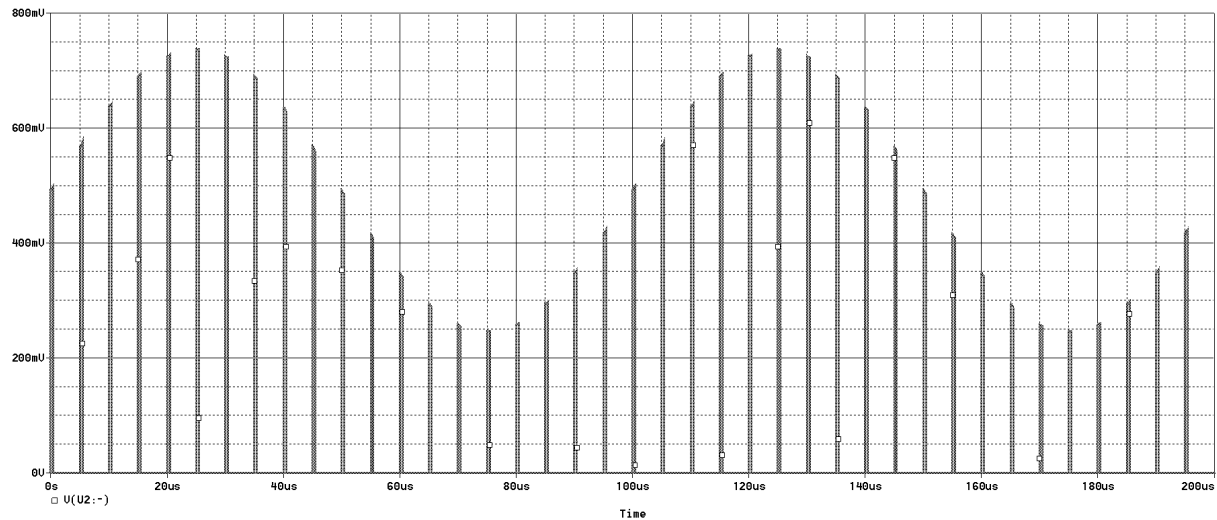
2. Modify the model parameters of the NMOS transistor with .model Mbreakn NMOS VTO=2 KP=0.05m W=50u L=0.1u



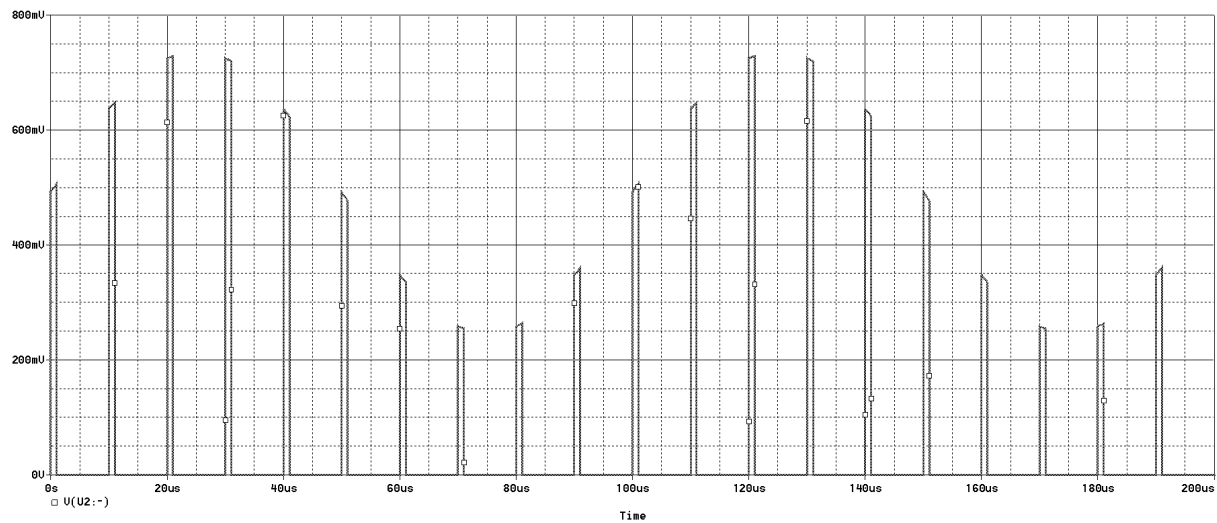
3. Setup a time-domain transient simulation in the simulation profile
Stop time: 0.2m
Step Size: 0.002m

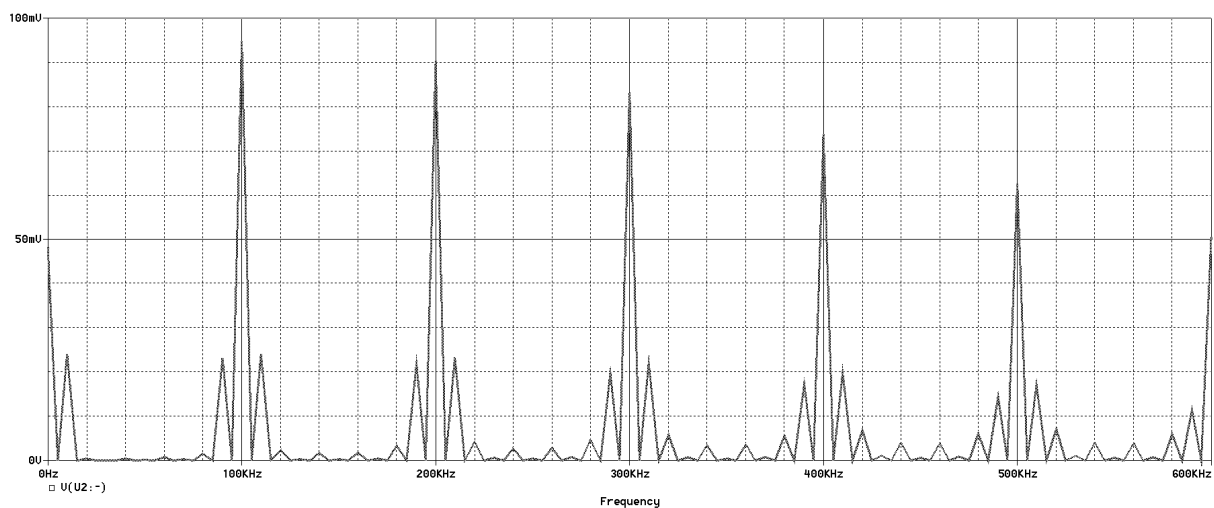


a. 200kHz Sampling Frequency

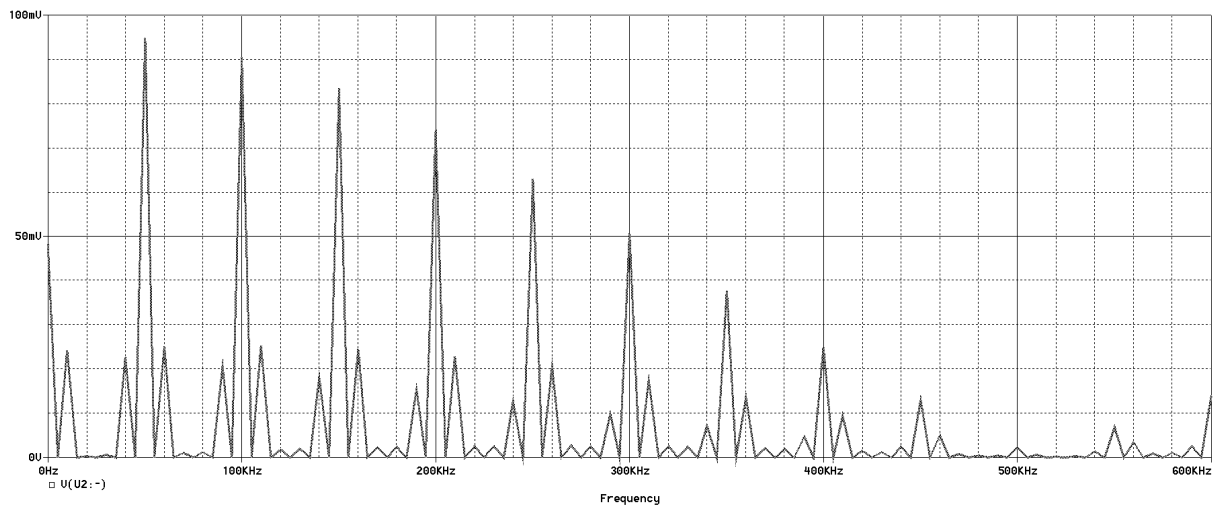
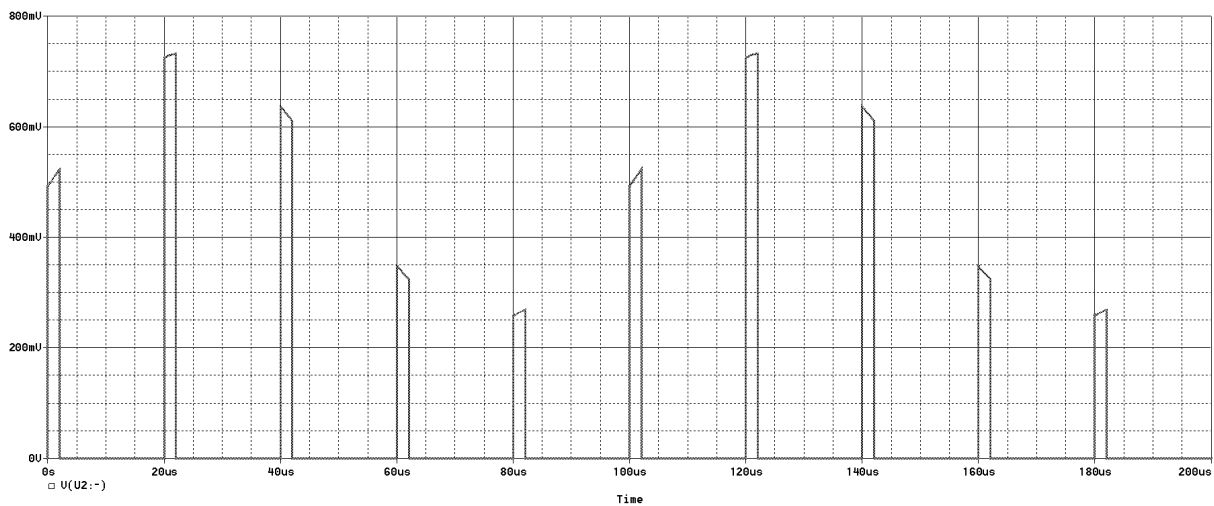


b. 100kHz Sampling Frequency

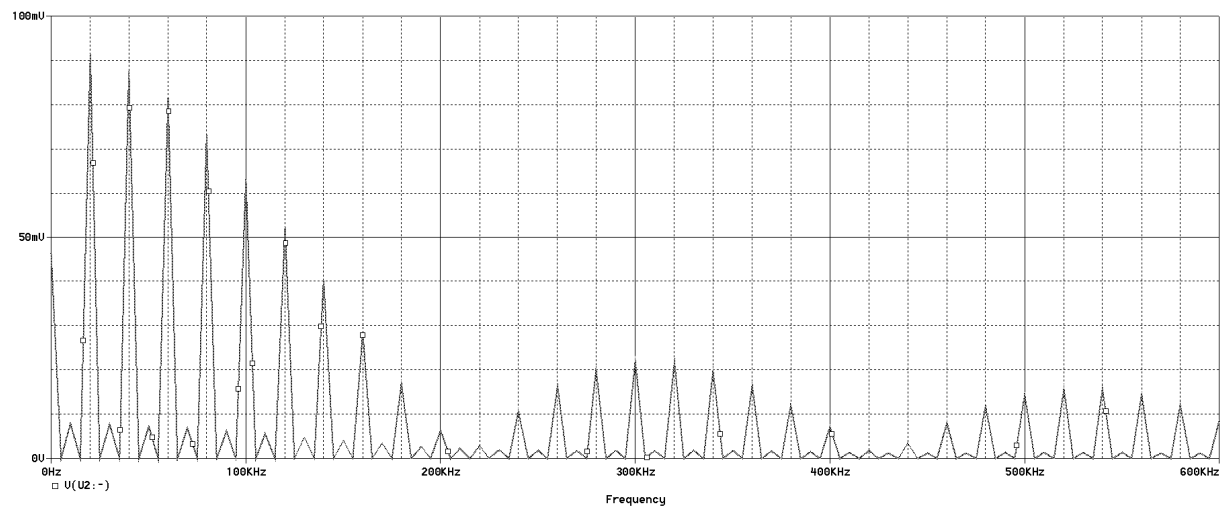
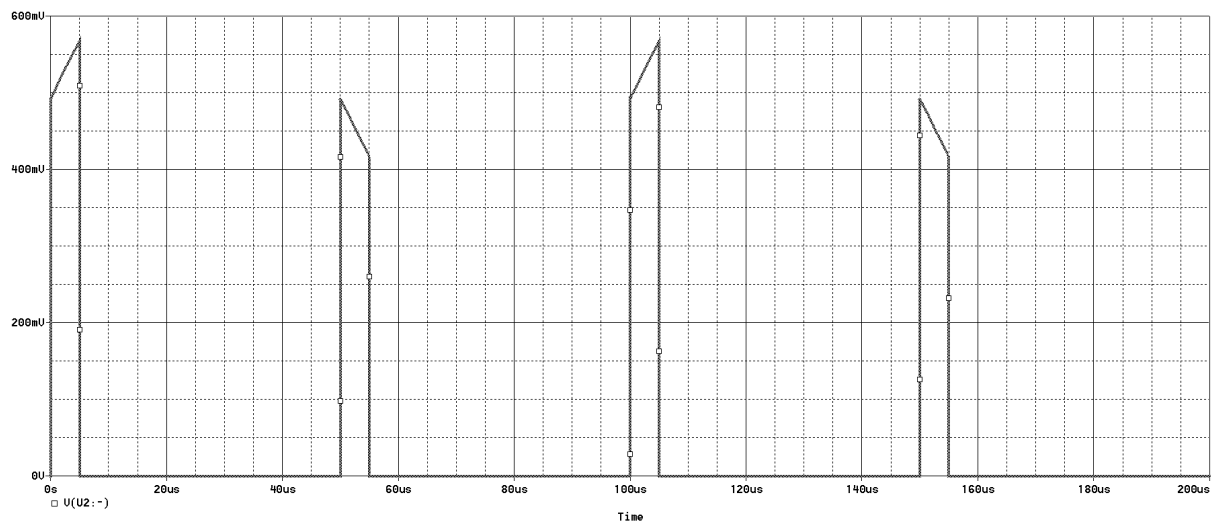




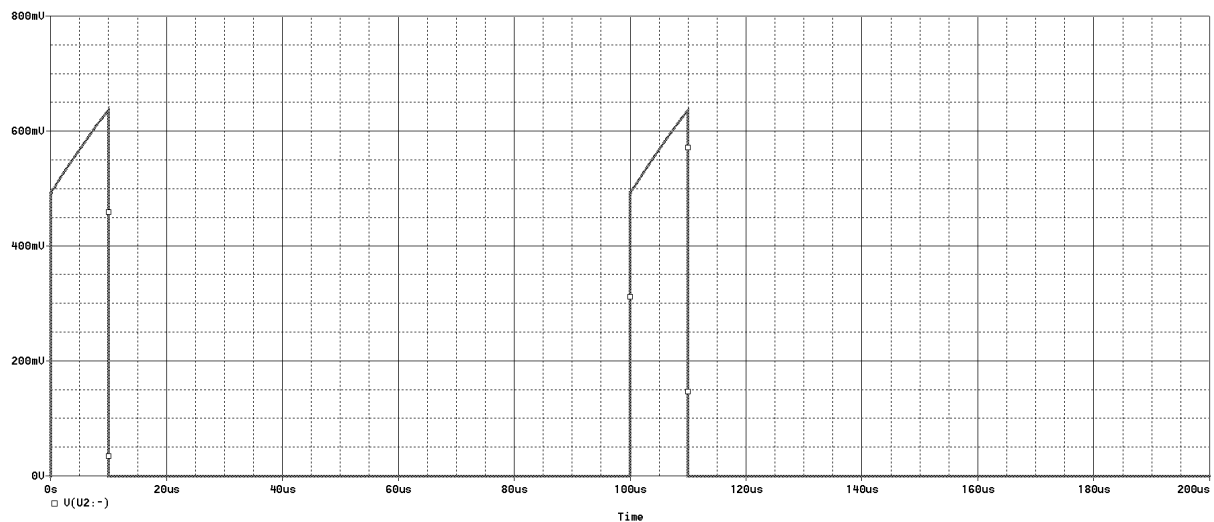
c. 50kHz Sampling Frequency

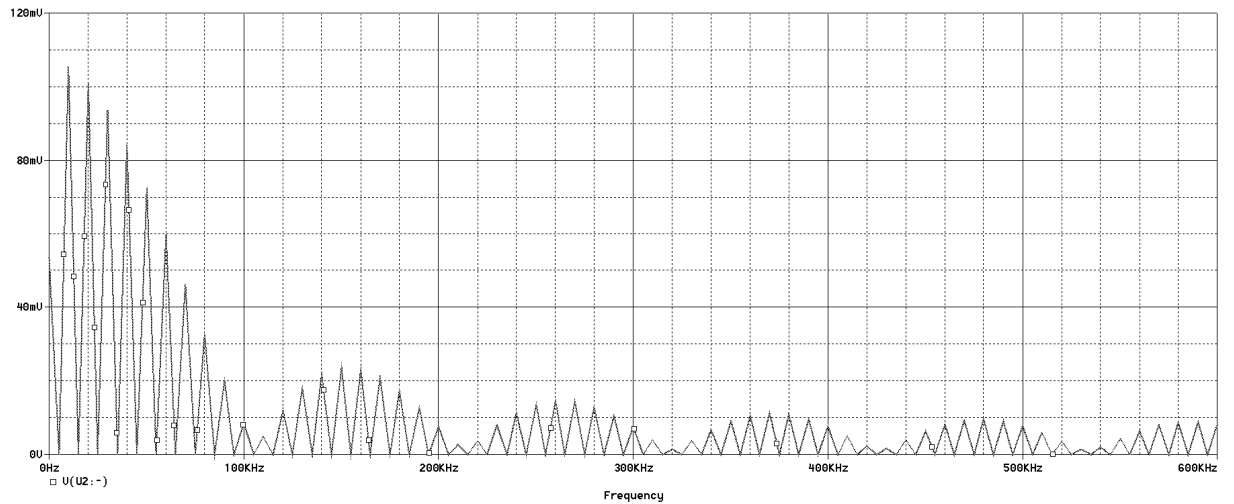


d. 20kHz Sampling Frequency

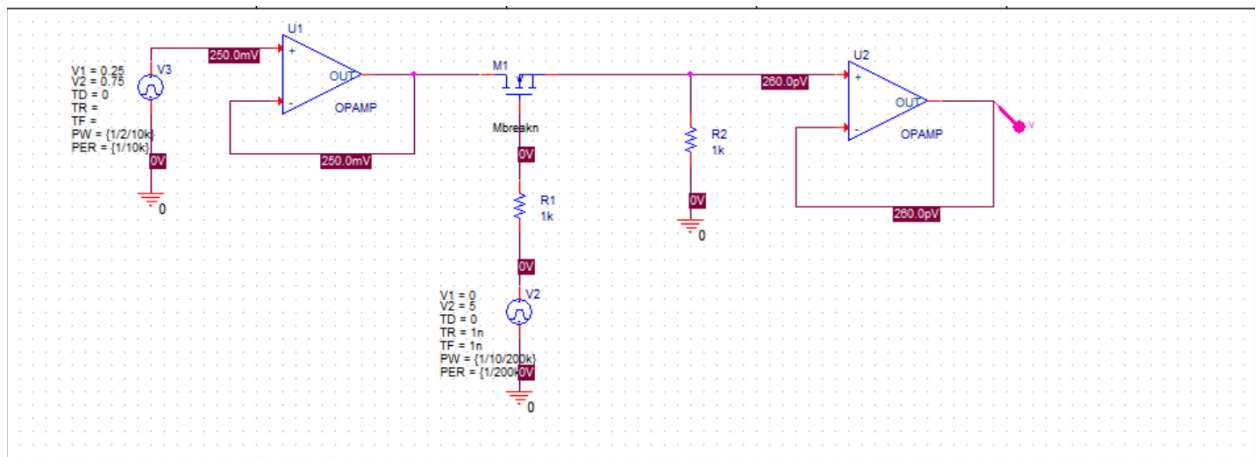


e. 10kHz Sampling Frequency

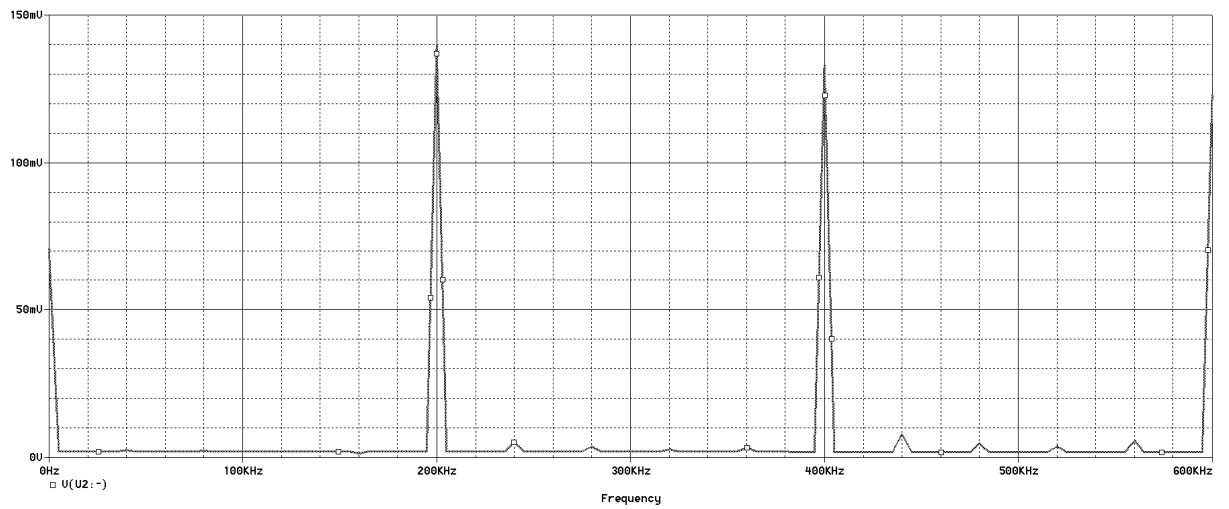
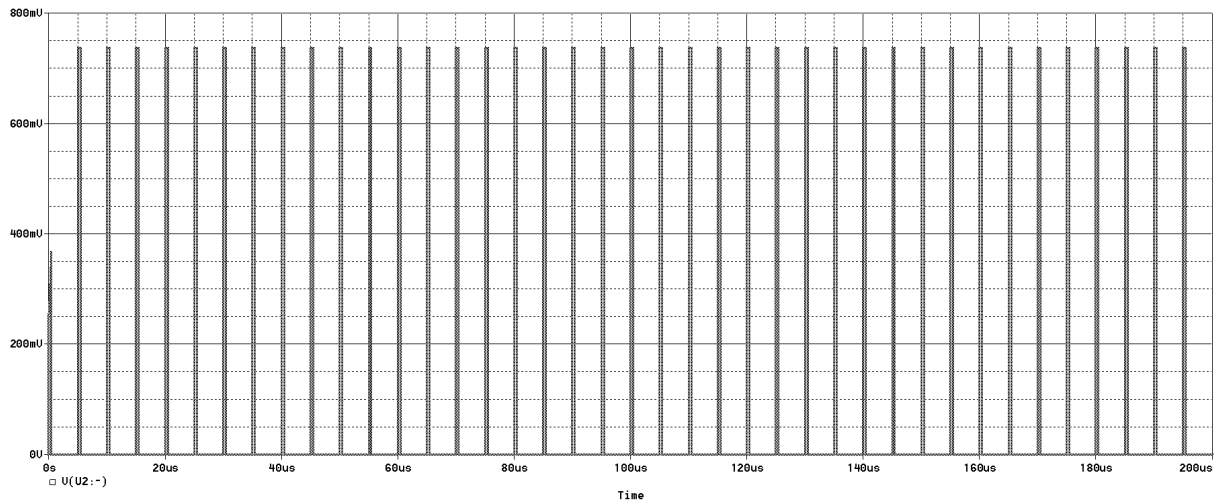




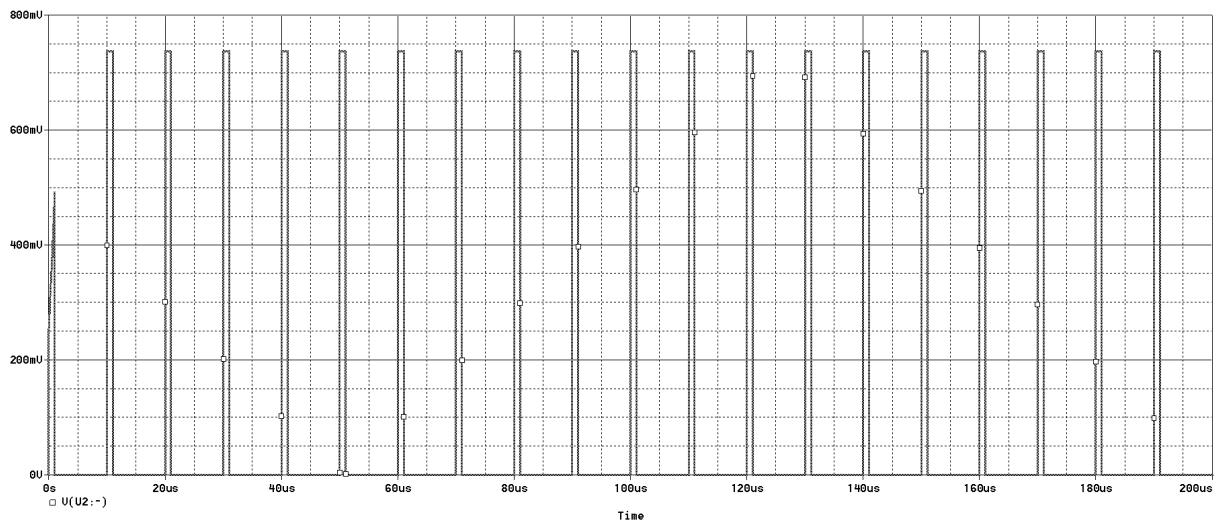
4. Change the analog input from a sine wave to a pulse waveform @ 10kHz, 50% duty cycle, +0.5 VDC offset, 0.5Vpp
 “Pulse” voltage source V1 = 0.25, V2 = 0.75, PW = {1/2/10k}, PER = {1/10k}
 Sampling frequency (f_s = 200k, 100k, 50k, 20k, 10k)

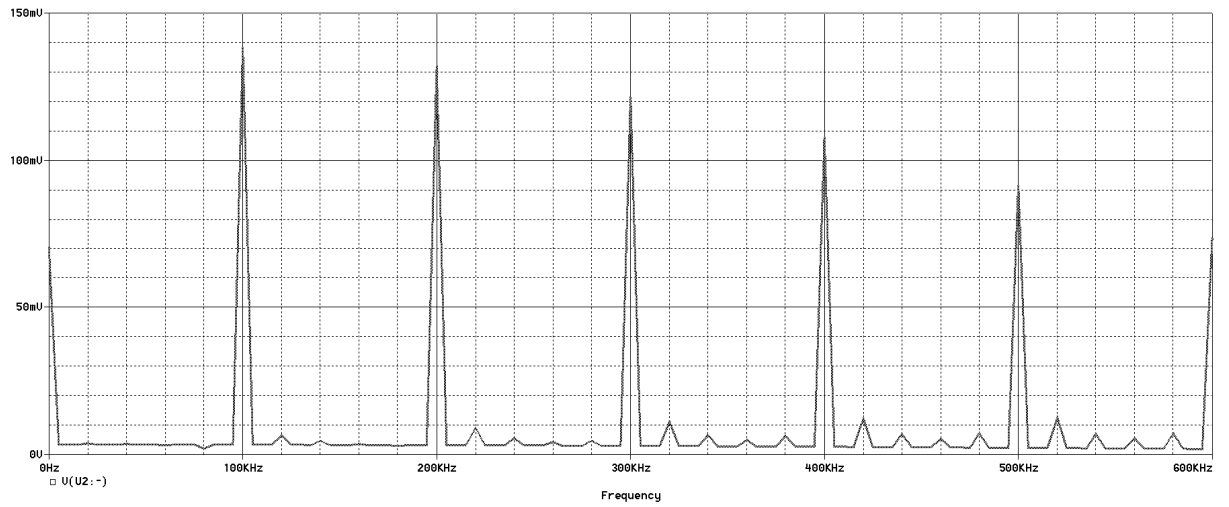


- a. 200kHz Sampling Frequency

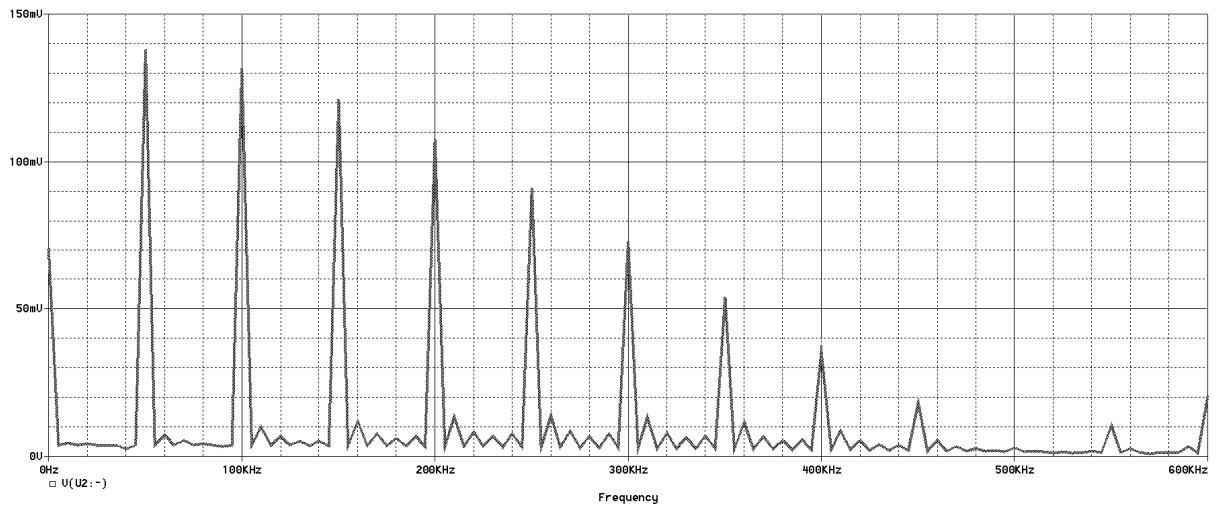
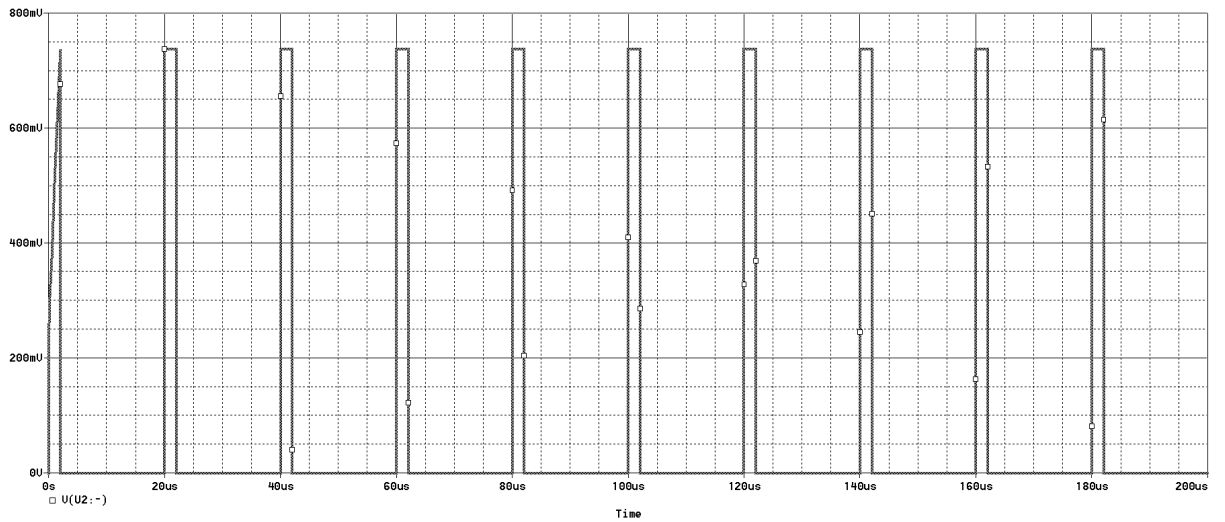


b. 100kHz Sampling Frequency

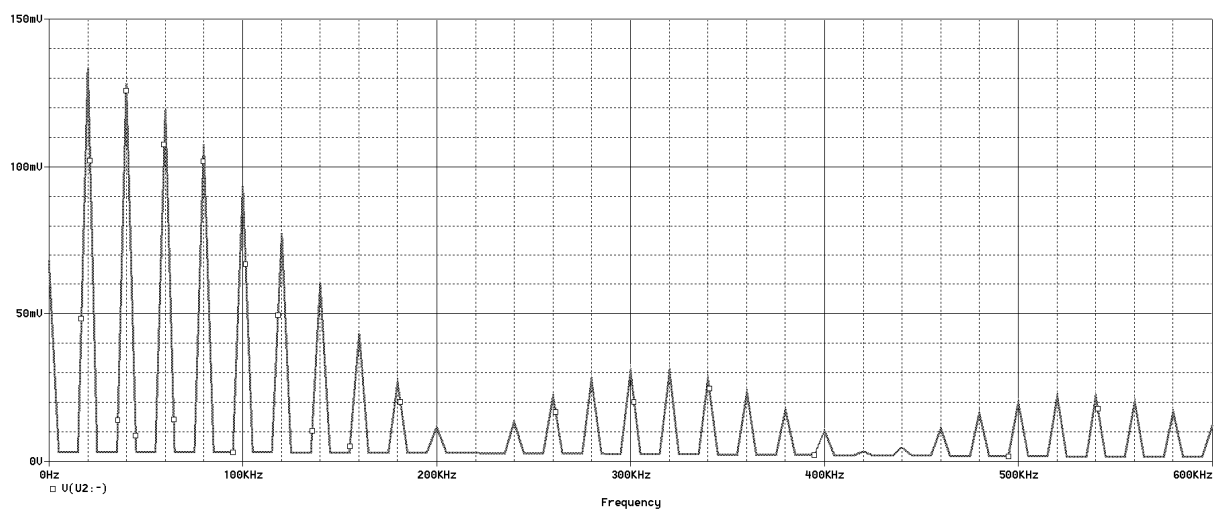
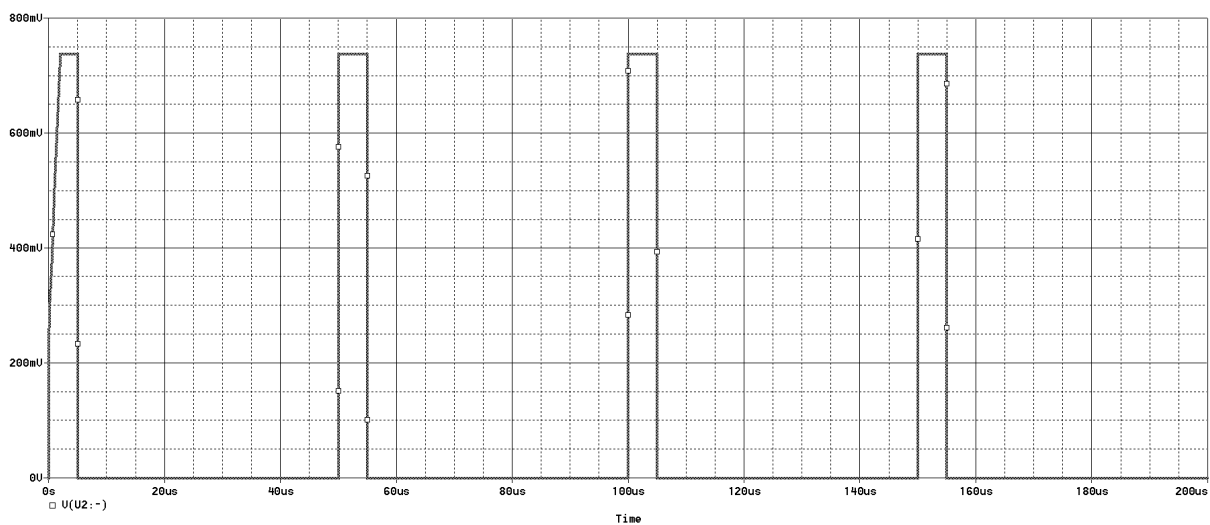




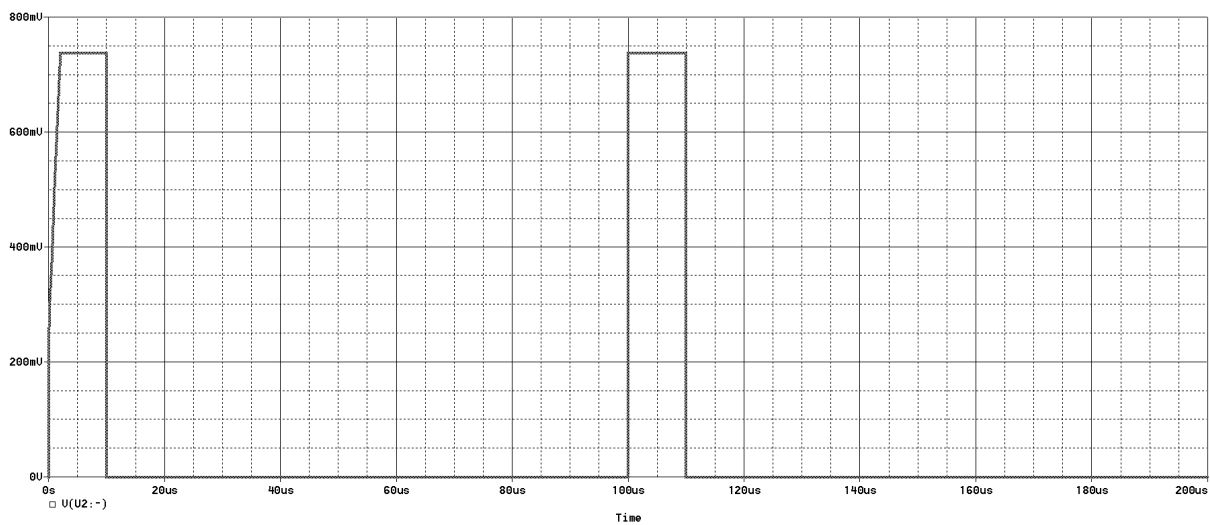
c. 50kHz Sampling Frequency

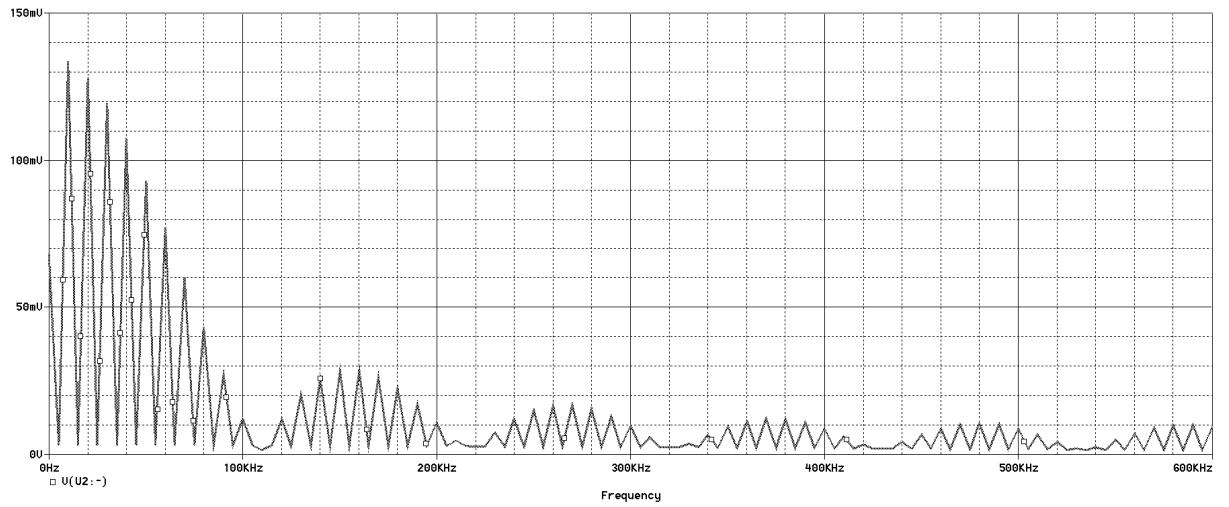


d. 20kHz Sampling Frequency

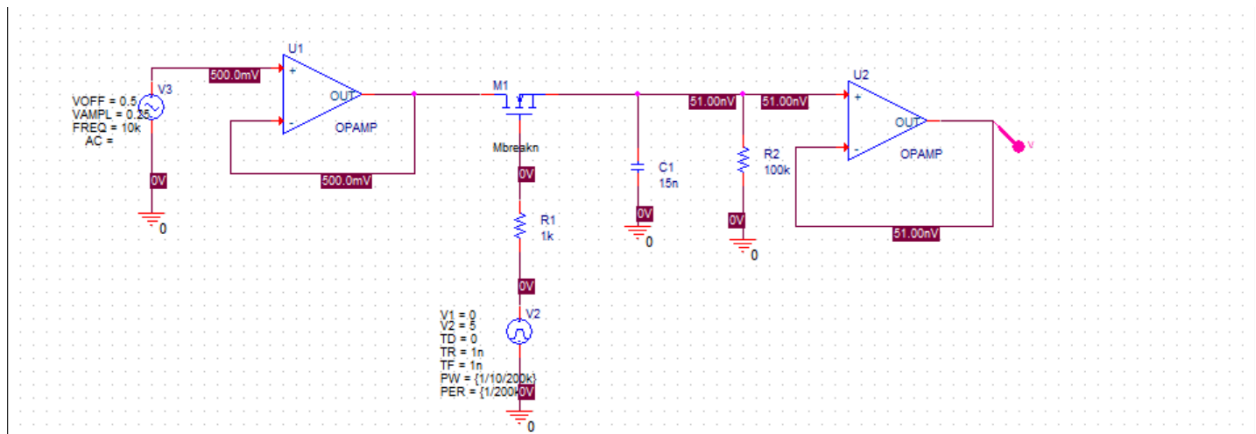


e. 10kHz Sampling Frequency

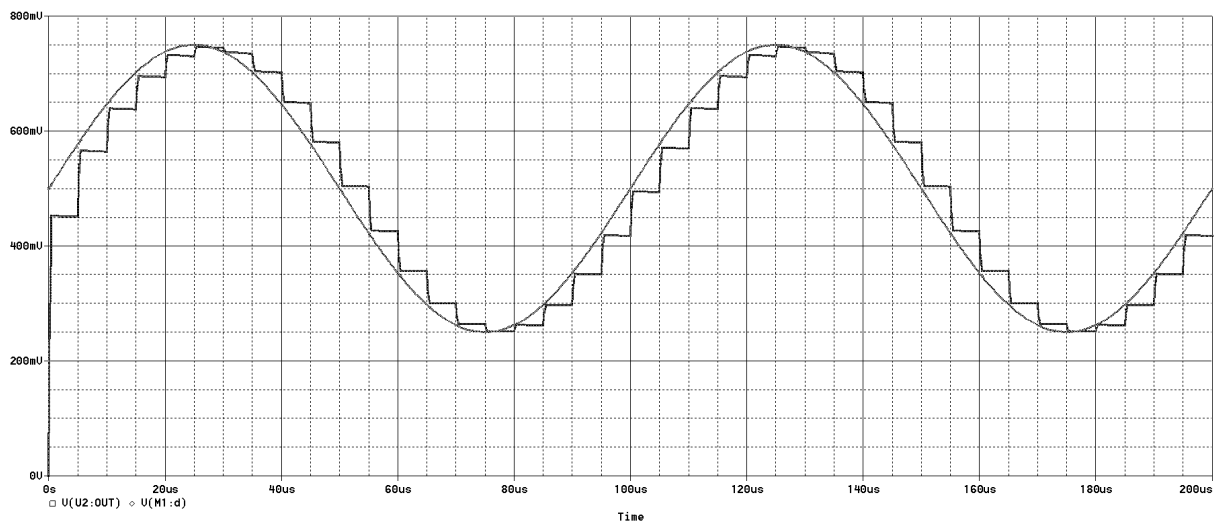




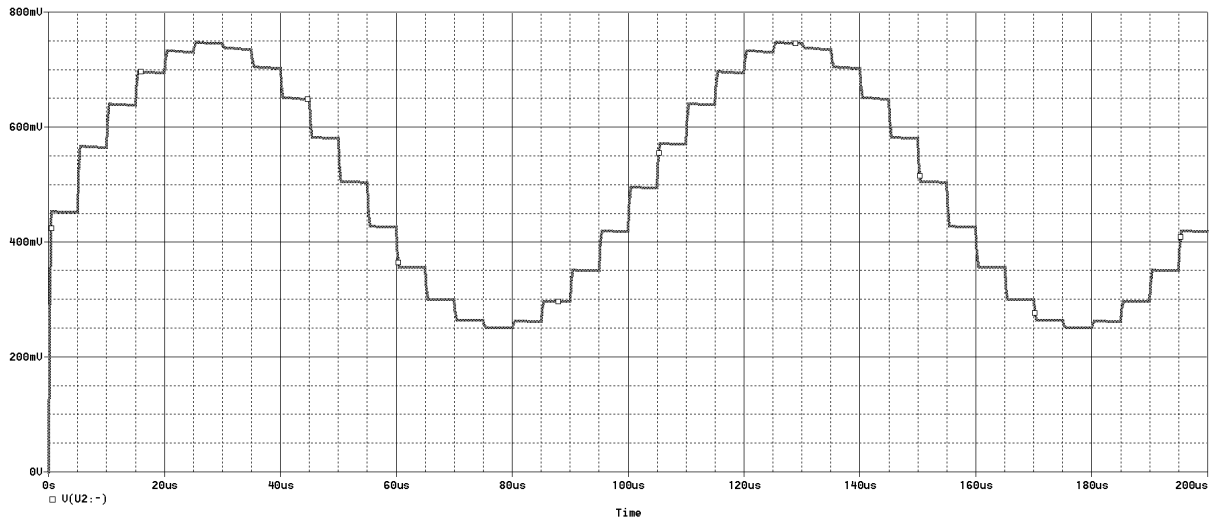
- Change the circuit adding a capacitor of 15nF and changing resistance of R2 to 100k



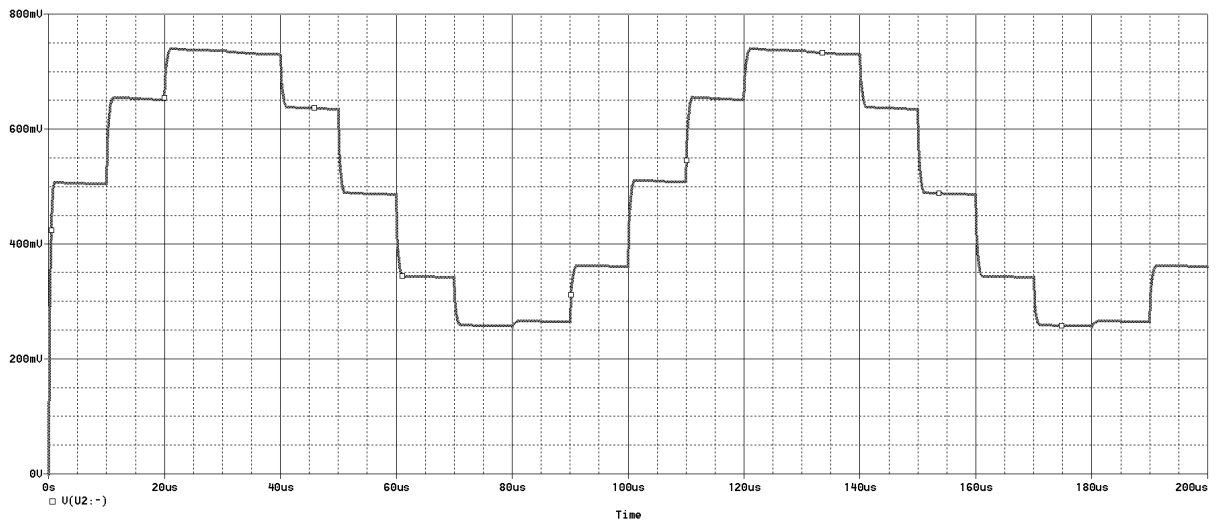
Source Frequency and Sampling Output (Time Domain):



a. 200kHz Sampling Frequency

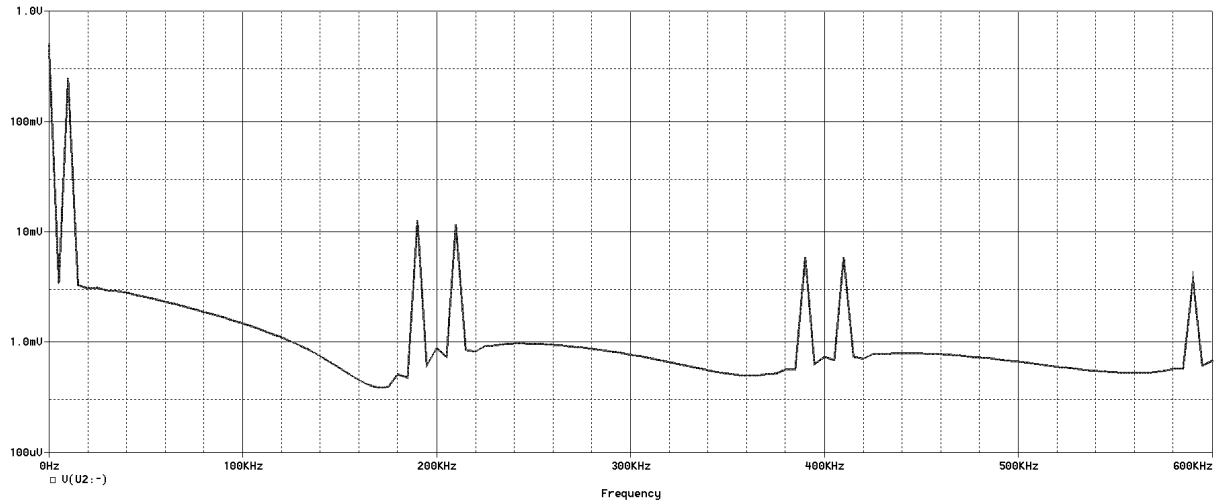


b. 100kHz Sampling Frequency

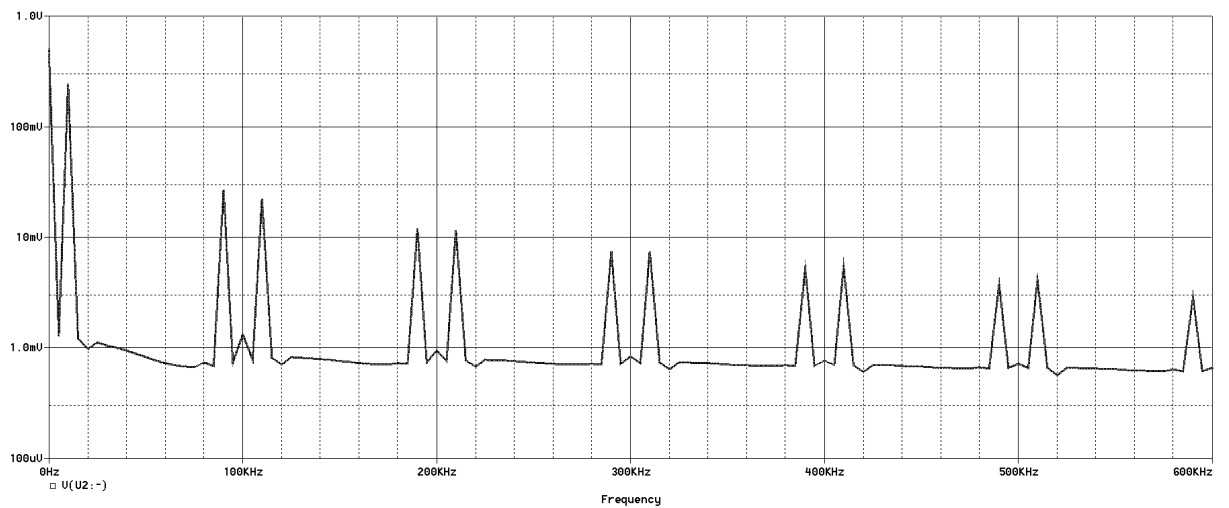


6. Record the time domain and FFT plots for $f_s = 200\text{kHz}$ and 100kHz . Change Y-axis setting from linear to log.

a. 200kHz Sampling Frequency



b. 100kHz Sampling Frequency



Conclusion:

The given sampling circuit allows adjustment to the sampling frequency through changes in the pulse width and period. An equation is set for the pulse width and period to easily test frequencies. The period is the inverse of the sampling frequency ($1/f_s$). Pulse width is the source period divided by the sampling frequency ($1/10\text{kHz}/f_s$). The source period is 0.1 millisecond, meaning a cycle will occur in that time frame. A requirement for the data was two cycles of the signal in frame, so the run time was 0.2 milliseconds ($2 \times 0.1\text{ms}$) and the maximum step size was one hundredth of 0.2 milliseconds ($0.2\text{ms}/100$). The frequency domain had to be manually adjusted to set the boundaries at 0Hz to 600kHz.

The first simulation went through 200kHz, 100kHz, 50kHz, 20kHz, and 10kHz. The highest frequency, 200kHz, had the most amount of pulses in each cycle and the most defined envelope compared to the lower frequencies. The shape in the time domain is based off of the sinusoidal source signal. 50kHz, 20kHz, and 10kHz all had one pulse in each cycle but the longest pulse width was the 10kHz. The 10kHz pulse width is measured at 10 microseconds, which does not compare to the 200kHz pulse width at 0.5 nanoseconds. The faster the pulse width, the more pulses are able to fit into the cycle. Frequency domain provides a different view, showing the lower frequencies having more frequency content. As the sampling frequency goes lower, the frequency domain becomes more defined and shaped similarly to a sinc function. The sidebands are present until it arrives at the lowest frequency, 10kHz, because it is now violating the Nyquist sampling rate requirement. This leads to a type of distortion called aliasing. The Nyquist sampling rate needs the sampling frequency to be twice the amount of the source frequency to be a valid sample.

The next section has a pulse waveform as the source signal. The results are similar to the sinusoidal signal with the source frequency increasing and leading to a decrease in frequency content. The time domain of the pulse waveform does not have an envelope or shape. The pulses have the same amplitude throughout its run time. The frequency domain has less evident sidebands than the sinusoidal signal, but demonstrates the same sinc behavior as the fundamental frequency decreases. Nulls are defined from 50kHz and below in both sinusoidal and pulse signals.

The last section presents the “holding” effect on the fundamental frequency. The term means to stall the signal at a voltage, therefore creating a step outline in the time domain. Only 100kHz and 200kHz are examined but give a good visual representation of the step outline. For example, the period for 100kHz is 10 microseconds. The voltage is held for about 10 microseconds, released, then increased or decreased in amplitude, and repeats until end of run time. The Fourier view needs the Y-axis to be adjusted to log to present the amplitude in terms of power. The harmonic amplitudes are now low and the sidebands have become larger than the harmonic amplitudes.